# Yifan Dong

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## Education

**Purdue University**, West Lafayette, IN — Aug 2023 – Present
*Ph.D. in Electrical and Computer Engineering*

- Research interest: Optimization and Machine Learning for Power Systems and Electricity Markets
- GPA: 3.84/4.0
- Coursework: Computational Method in Optimization, Convex Optimization, Computational Method for Power System Analysis, Optimization for Deep Learning, Algorithm Design, Analysis and Implementation, etc.

**North China Electric Power University**, Beijing, China — Aug 2019 – June 2023
*B.Eng. in Electrical Engineering and Automation*

- GPA: 4.31/5.0 (Top 1%)
- Coursework: Power System Analysis, Power System Economy and Management, Automatic Control Theory, etc.

## Experiences

**Independent System Operator - New England**, Holyoke, MA — June 2026 – Aug 2026
Student Intern at Advanced Technology Solutions (ATS)
*Mentors:* Dr. Feng Zhao, Dr. Xiaochu Wang, and Dr. Yufan Zhang

- *Project 1: Bidding Behavior in the Day-Ahead Ancillary Services (A/S) Market*
  - Conducted **theoretical and statistical analyses of A/S offers**, characterizing offer-price patterns, evaluating offer competitiveness, and examining relationships between offers and award outcomes.
  - Developed a **data-driven framework for unit classification** that learns low-dimensional representations of daily bidding behavior, identifies recurring A/S offer patterns through unsupervised clustering, and groups units by their recurring bidding profiles.
- *Project 2: In-House Probabilistic Evaluation Tool for Resource Adequacy (PETRA) Development*
  - Evaluated **input consistency between PETRA and GE MARS**, identifying and resolving conversion discrepancies to establish a reliable basis for subsequent output validation.
  - Validated **PETRA's resource-adequacy modeling and dispatch logic** using analytical benchmarks and production-scale test cases, examining loss-of-load behavior, energy-storage dispatch, load-shedding strategies, and multi-scenario reliability outcomes.

**ASPIRE Research Center**, Purdue University, West Lafayette, IN — Aug 2023 – Present
Research Assistant | *Advisor:* Prof. Junjie Qin

- *Topic 1: Privacy-Preserving Demand Flexibility Aggregation for Flexible Loads*
  - Designed a **privacy-preserving framework for aggregating demand flexibility**, enabling large-scale coordination among devices while keeping user data local and secure.
  - Re-architected a bilevel optimization problem into a **scalable single-level learning framework**, enabling efficient end-to-end optimization for large-scale flexibility aggregation.
  - Developed a **federated gradient-based algorithm** with backend logic for secure exchange of anonymized model parameters, reducing communication overhead and preserving data privacy.
  - Conducted extensive testing and performance evaluation, **demonstrating significant improvements in system efficiency and aggregate flexibility** over existing baselines.
- *Topic 2: Real-Time Charging Optimization for Electric Roadways*
  - Designed and implemented a **real-time charging control system** for electric roadways, enabling safe, efficient wireless power transfer between roadway microgrids and electric vehicles under dynamic traffic conditions.
  - Re-architected the charging-control problem using **dynamic programming (DP)**, exposing **state-transition structure and constraints** that enabled a simplified and implementable solution strategy.
  - Developed a **causal, online scheduling algorithm** that exploits the DP-derived properties to support **lightweight, real-time decisions** without relying on future information.
  - Evaluated the system across diverse traffic scenarios, demonstrating **low suboptimality, reliable constraint satisfaction, and strong robustness to prediction errors**.

**Purdue ECE**, West Lafayette, IN Fall 2024, Spring 2026
Teaching Assistant for ECE 31032 - Power Systems Engineering

- Supported instruction on **power flow, transmission lines, transformers, and economic dispatch.**
- Held weekly office hours, designed assignments and exams, and graded projects.

## Publications

[1] **Yifan Dong**, Ge Chen, and Junjie Qin, "Federated Aggregation of Polyhedral Demand Flexibility Sets", *IEEE Transactions on Smart Grid*, 2026. [Paper]

[2] **Yifan Dong**, Ge Chen, Junjie Qin, S. Sivaranjani, Xiaonan Lu, Dionysios Aliprantis, and David Love, "Real-Time Charging Control for Electric Roadways", *IEEE Transactions on Smart Grid*, 2026. [Paper]

[3] **Yifan Dong**, Junjie Qin, S. Sivaranjani, Xiaonan Lu, Dionysios Aliprantis, and David Love, "Real-Time Charging Control for Electric Roadways: Formulation and Causal Algorithms", *IEEE Power & Energy Society General Meeting*, 2024. [Paper]

## Presentations

[1] "*Federated Aggregation of Demand Flexibility*", 2025 ASPIRE Annual Meeting, Logan, UT [Poster]

[2] "*Real-Time Charging Control for Electric Roadways*", 2024 ASPIRE Annual Meeting, Logan, UT [Poster]

[3] "*Real-Time Charging Control for Electric Roadways*", 2024 IEEE PES General Meeting, Seattle, WA [Poster]

## Technical Skills

Python, SQL, MATLAB, PyTorch, scikit-learn, Pandas, NumPy, CVXPY, Gurobi, CPLEX, SUMO